

Unit 1.2 Absolute value

Note Title

2015/01/26

p A6 - A9

Notation:

$|a|$ — distance on the real line from a to 0.
— always positive (or 0)

Eg.

$$|-4| = 4$$

$$|-\frac{7}{2}| = \frac{7}{2}$$

$$|5| = 5$$

$$|-\pi| = \pi$$

$$|0| = 0$$

$$|10^4| = 10^4$$

Question: Is $-a$ a negative number?

$$a = 5 \Rightarrow -a = -5 < 0$$

$$a = -5 \Rightarrow -a = -(-5) = 5 > 0$$

NO,
depends on
 a

Definition:

$$|a| = \begin{cases} a & \text{if } a \geq 0 \\ -a & \text{if } a < 0 \end{cases}$$

$$a = -3 \Rightarrow |a| = -a = -(-3) = 3$$

$$a = 4 \Rightarrow |a| = 4$$

Eg $|-5| = 5$

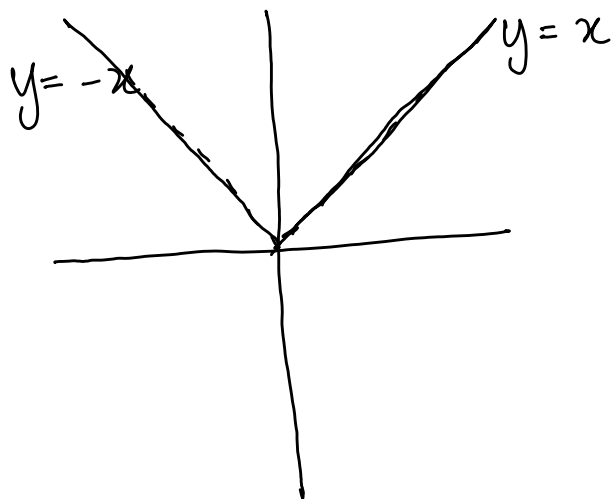
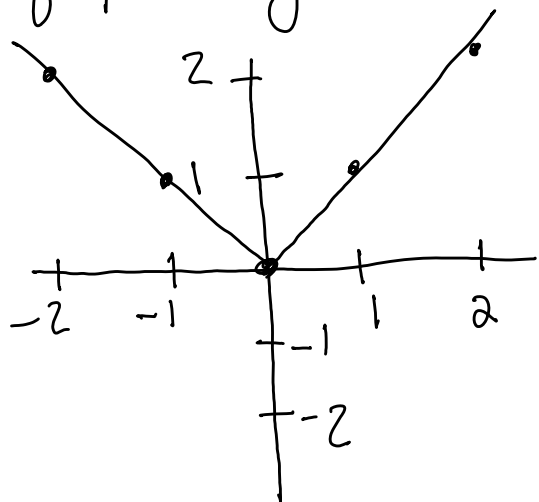
$$|6| = 6$$

$$|-\frac{1}{2}| = -(-\frac{1}{2}) = \frac{1}{2}$$

In terms of x :

$$y = |x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$$

Graphically:



Properties of absolute value:

1. $|xy| = |x||y|$

Eg $x = -2, y = 4 \Rightarrow |xy| = |-8| = 8$
and $|x||y| = |-2||4| = 8$

2. $\left| \frac{x}{y} \right| = \frac{|x|}{|y|}$

, $y \neq 0$.

3. $|x^n| = |x|^n$

$\underbrace{|(-3)^3|}_{27} = | -3 |^3_{27}$

4. $\sqrt{x^2} = |x|$ NB.

positive $\rightarrow$

Eg. $\sqrt{(-4)^2} = 4$

5. $|x| \geq 0$ for all $x \in \mathbb{R}$

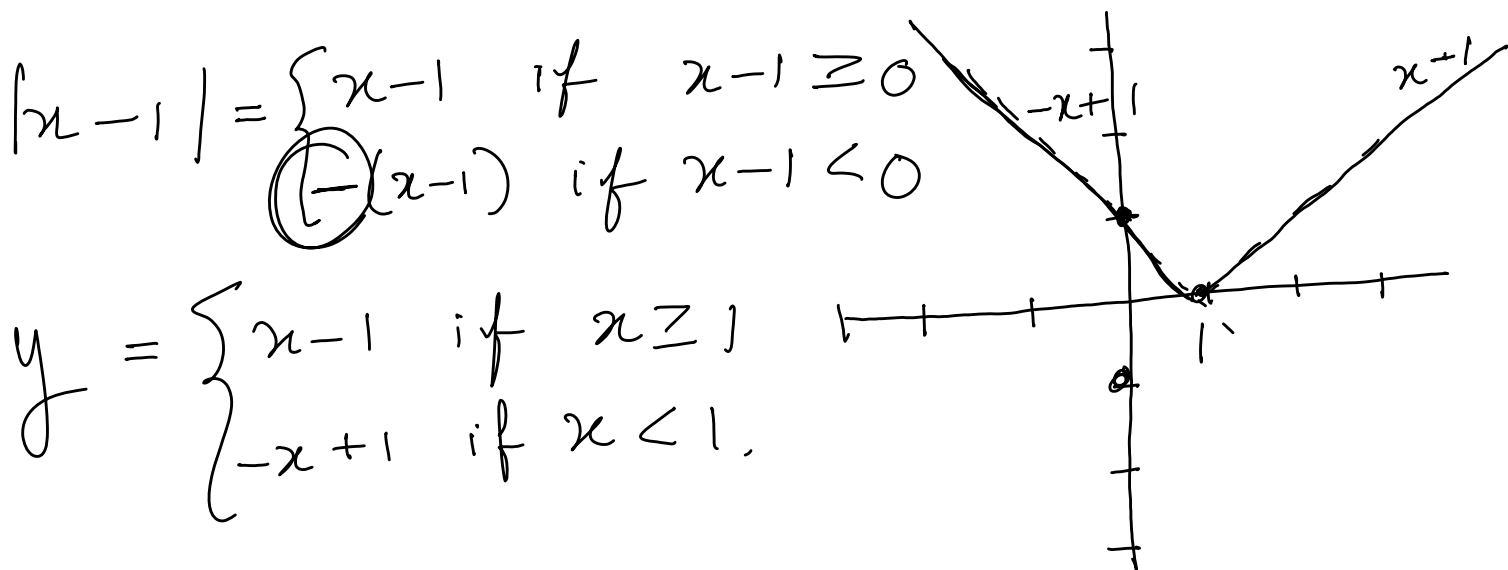
6. $|x| = 0$ if and only if $x = 0$

$$\left. \begin{array}{l} |x| = 0 \Rightarrow x = 0 \\ |x| = 0 \Leftarrow x = 0 \end{array} \right\} |x| = 0 \Leftrightarrow x = 0$$

if and only if.

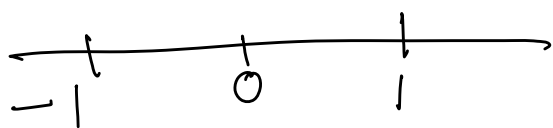
Other graphs:

1. $f(x) = |x - 1|$



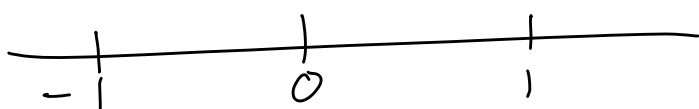
2. In preparation for this example! NO!!

Solve $x^2 \leq 1 \Rightarrow \textcircled{x \not\leq \pm 1}$



$$-1 \leq x \leq 1$$

Solve: $x^2 > 1$

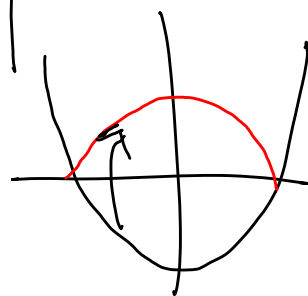
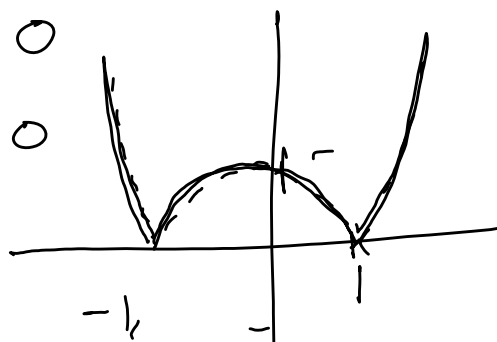


$$\Rightarrow x > 1 \text{ or } x < -1$$

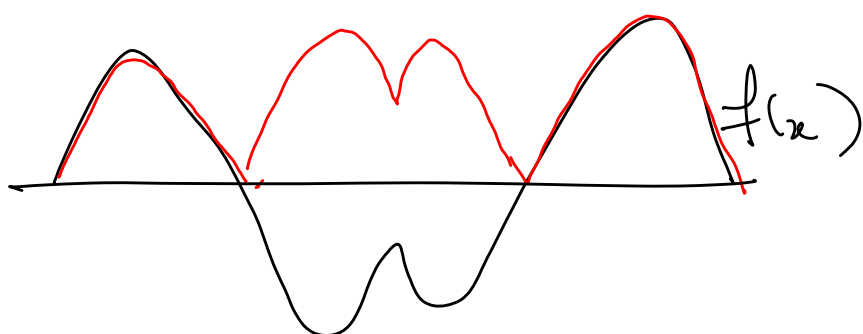
Eg. Sketch $f(x) = |x^2 - 1|$

$$|x^2 - 1| = \begin{cases} x^2 - 1 & \text{if } x^2 - 1 \geq 0 \\ -(x^2 - 1) & \text{if } x^2 - 1 < 0 \end{cases}$$

$$= \begin{cases} x^2 - 1 & \text{if } x^2 \geq 1 \\ -x^2 + 1 & \text{if } x^2 < 1 \end{cases}$$

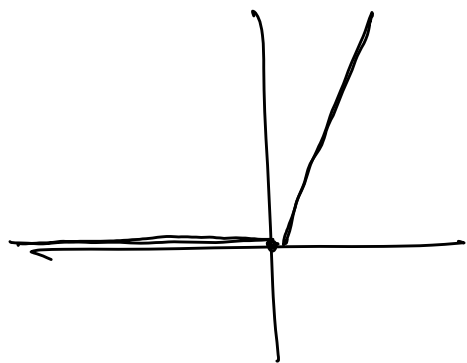


$$y = \begin{cases} x^2 - 1 & \text{if } x \geq 1 \text{ or } x \leq -1 \\ -x^2 + 1 & \text{if } -1 < x < 1 \end{cases}$$



$$|f(x)|$$

Eg. Sketch $f(x) = |x| + x$



$$= \begin{cases} x + x & \text{if } x \geq 0 \\ -x + x & \text{if } x < 0 \end{cases}$$

$$= \begin{cases} 2x & \text{if } x \geq 0 \\ 0 & \text{if } x < 0 \end{cases}$$

Theorem for solving equations involving absolute value:
(p A7 16)

Suppose $a > 0$:

NB $\left\{ \begin{array}{l} 1. |x| = a \iff x = \pm a \\ 2. |x| < a \iff -a < x < a \\ 3. |x| > a \iff x > a \text{ or } x < -a. \end{array} \right.$

Examples : Solve:

1. $|x| < -3 \Rightarrow x \in \emptyset$

2. $|x| = 4 \Rightarrow x = \pm 4$

If $a < 0$
then
THINK.

3. $|x^2 - 14| = 1 \Rightarrow x^2 - 14 = 1 \text{ or } x^2 - 14 = -1$
 $\Rightarrow x = \pm\sqrt{15} \text{ or } x = \pm\sqrt{13}$

4. $|x - 3| < 4 \Rightarrow -4 < x - 3 < 4$
 $\Rightarrow -1 < x < 7$
 $\Rightarrow x \in (-1, 7)$

5. $|x - 4| > 6 \Rightarrow x - 4 > 6 \text{ or } x - 4 < -6$
 $\Rightarrow x > 10 \text{ or } x < -2$
 $\Rightarrow x \in (-\infty, -2) \cup (10, \infty)$

More examples:

$$6. |x^2 - 1| = 0 \Rightarrow x^2 - 1 = 0 \Rightarrow x^2 = 1 \\ \Rightarrow x = \pm 1$$

$$7. |1 - x| \leq 2 \Rightarrow -2 \leq 1 - x \leq 2 \\ \Rightarrow -3 \leq -x \leq 1 \\ \Rightarrow 3 \geq x \geq -1 \Rightarrow -1 \leq x \leq 3 \\ \Rightarrow x \in [-1, 3]$$

$$8. |2 - 3x| > 1 \\ \Rightarrow 2 - 3x > 1 \quad \text{or} \quad 2 - 3x < -1 \\ \Rightarrow x < 1/3 \quad \text{or} \quad x > 1 \Rightarrow x \in (-\infty, 1/3) \cup (1, \infty)$$

Examples (requiring more thinking)

$$1. |2 - x| > 0 \Rightarrow 2 - x \neq 0 \\ \Rightarrow x \neq 2 \\ \Rightarrow x \in \mathbb{R} \setminus \{2\}$$

$$2. \frac{x}{|x-2|} > 0^+ \Rightarrow x > 0, \quad x \neq 2 \\ \Rightarrow x \in (0, \infty) \setminus \{2\}$$

$$3. \frac{|x|}{|x-1|} \geq 0 \Rightarrow x \in \mathbb{R} \setminus \{1\}$$

$$4. |x-1| > -4 \Rightarrow x \in \mathbb{R}$$

$$5. |x-1| < -4 \Rightarrow x \in \emptyset$$